Decoys That Do Work: Evaluating Executed Cover for LLM Agent Communication Privacy

***Abstract*—Decoy tool calls change both an LLM Agent's encrypted traffic and the work its services perform. We study how call generation affects task recognition and evaluate inference, goal completion, business-state integrity and cost on the same executions. In 3,000 single-host executions with two fixed-weight planners over MCP and TLS 1.3/HTTP/2, ordinary tool calls follow a model decision, whereas controller-inserted cover bypasses that wait. With supplied call windows, label-free timing removal raises recognition from 75 to 112 and from 80 to 108 of 120 tasks. Waits sampled from training traces defeat this shortcut, but attacks using destinations, ordering, sizes and timing recognize 93–106/120; three of four comparisons with selected size-only attacks survive Holm correction. Without supplied call windows, the prespecified removal pipeline loses its advantage, but its idle-gap events rarely match true calls, so this diagnoses the pipeline rather than showing that withheld boundaries protect; recognition remains above chance. A complete shadow task, which also adds a second business action to every episode, reduces the strongest tested call-window counts to 70/120 and 64/120 at about twice the model tokens. These measurements distinguish an exploitable dispatch shortcut from residual trace context and quantify complete-policy costs. A four-axis design space and joint outcome record connect the traffic results to the service capabilities and business-state authority each policy requires.**

1. Introduction

An LLM Agent receives a support ticket, checks an order and submits a refund. An observer of the encrypted path to the commerce service sees contacted endpoints, exchange sizes and waiting times. Recent Agent studies use such metadata to infer prompts, tasks, workflows and user traits [2], [3], [4], [9]. Adding decoy calls can obscure the requested action, but each extra exchange must be generated somewhere and terminated somewhere. These choices determine its traffic signature and business cost.

The mechanism. In the planner loop we measure, an ordinary tool call follows a model decision. A controller can insert a decoy directly after a tool response, bypassing the next model wait. That shortcut separates the decoy by its inter-call gap. Delaying the call changes this feature, but leaves its destination, position, size and relationship to the other calls available to an attacker. A complete shadow task changes those relationships as well as adding a second planner's work. We compare these implemented choices on actual TLS records and track the work performed with each trace.

Classical query obfuscation already links decoy quality to its generator [63], [64], [65]. Our question is how that principle interacts with an Agent's model decisions and tool authority. Immediate cover is therefore a calibration arm, not a proposed defense: cover that ignores the honest timing law is known to be separable [69], [64], and we include it to confirm that the attack suite detects this in a live Agent loop. A byte-only attack can miss the controller's timing shortcut; a timing-only explanation can miss remaining call structure; a goal-only check can miss collateral business effects. The measurements therefore pair observation information with inference, task outcome, business state and resource use.

Setting. We compare four cover policies and no cover in 3,000 live executions of Qwen3:8B and Llama3.1:8B over MCP and TLS 1.3/HTTP/2 on one host. The provider cooperates by padding responses and permitting isolated execution; history-based policies also use server-side minimum holds. Section 4 states these capabilities. When only request shape is private, protocol padding may suffice. Hiding response activity calls for additional responses, which a cooperating provider can synthesize or an application call can produce. Reaching the application need not require a business write.

Findings.

- **A controller shortcut is exploitable when calls are delimited.** Label-free gap removal raises recognition from 75 to 112 (Qwen) and from 80 to 108 (Llama) of 120 tasks; both gains survive Holm correction and reproduce on independent packet-capture clocks. With sampled waits, mixed-context attacks recognize 93–106/120. Three history-arm comparisons survive Holm correction; Qwen sequential does not ($p = .0559$).
- **The tested record-only pipeline does not establish a boundary effect.** Without call windows, selected removal recognizes 69/120 and 86/120 against baselines of 80 and 90. Its idle-gap events match true calls with IoU-.5 F1 of zero in eight of ten cells, and the suite lacks the stronger context and joint decoders, so this result diagnoses the pipeline rather than measuring what call boundaries are worth.
- **Shadow cover changes a complete policy's inference–cost trade-off.** The strongest tested call-window attack scores 70/120 and 64/120 at about twice the model tokens, with more and different

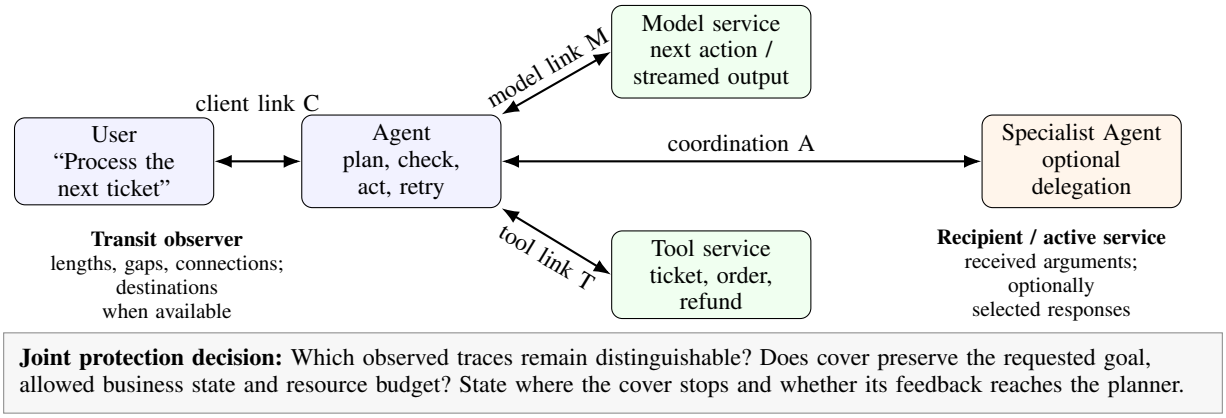


Figure 1. Running support-ticket example and observation rights. A tool result returns to the Agent and can enter the next model request. Each encrypted link exposes its own traffic shape; a recipient additionally sees the request addressed to it. Concurrent tickets may share a connection on either link. The arrows represent application communication, not a one-to-one mapping from tool events to packets.

cover calls. Shadow is also the only tested policy that adds a second business action to every episode; controller schedules leave the requested action last in 79 of 120 test fixtures, so the comparison does not isolate online planning. The main study records task and state outcomes on those same executions; a separate stress control illustrates collateral effects missed by goal completion.

Contributions. We contribute (1) a live measurement distinguishing a short-gap dispatch shortcut from residual trace context; (2) an observer-explicit comparison of implemented controller and shadow policies with matched inference, task, state and resource outcomes; and (3) an execution-aware evaluation method that states cover termination, state and feedback authority, and service support. Section 5.3 provides an ideal candidate-label reference for interpreting shadow results. Appendix A supplies the forty-four-setting background, rather than a field-wide evaluation-quality estimate.

2. Background and Motivation

2.1. Agent communication

An Agent uses an LLM to choose actions, and external feedback can change its later actions or termination. Figure 1 shows the links of the running example: client–Agent, Agent–model, Agent–tool and inter-agent coordination. A tool result returns to the Agent and can enter the next model request. Tools are commonly reached through the Model Context Protocol (MCP), either over local stdio or over Streamable HTTP [23]. Remote connections typically use TLS 1.3 and HTTP/2, which encrypt content but expose record lengths, directions and timing, and multiplex several request streams on one connection [25], [24]. A captured packet, TLS record, HTTP stream, tool call and complete task are therefore different units, and a boundary supplied by an evaluator is prior information, not an attacker output.

2.2. Communication side channels in Agents

A small but growing literature shows that Agent traffic leaks. Transit observers recover research prompts and user traits from contacted domains and timing [2], workflow classes from inter-agent communication graphs [3], and tasks or misuse from model-link and egress traffic [4], [9], [1]. Contacted recipients fingerprint Agent clients and models from protocol headers, terminal commands and tool-call arguments [6], [7], [48], [47], [35], and active defenders plant lures for Agent adversaries [10], [51]. The companion matrix maps all forty-four settings from fifteen such studies by observer access, target, unit and intervention. Most evaluate recognition; few evaluate a protection.

2.3. How traffic defenses are evaluated

Website-fingerprinting (WF) work has established what a defense evaluation must contain: realistic observer information and splits, attacks adapted to the protected distribution, and overheads measured on the same traffic [15], [16], [32]. Adaptive-attack and ML-security guidance make the same demands [19], [52]. Classical defenses pad, pace or add dummy traffic, and their cover never reaches an application [17], [18], [66], [34]. State-aware Agent benchmarks check environment outcomes [60], [61] but do not measure traffic.

2.4. Motivation for joint outcome records

Table 1 applies those requirements to the four Agent query/workflow privacy settings we located. WRA’s five research decoys lower query recovery from .770 to .565, but its report-utility scores of 8.55 and 8.54 belong to a different intervention that changes destination visibility [2]. The communication-graph study refits its attacker on live OHTTP captures and reports about thirteen-fold bytes, but no original-task outcome under that intervention [3]. No setting reports both an adapted attacker and matched utility,

and the only executed-cover setting reports no matched utility at all. This is a statement about four settings from two sources, not a field-wide frequency; it shows that the measurements a deployment decision needs have not yet been made together.

3. Threat Model

Assets and adversary. The private target is the user's requested task, here one of six support actions. The defender controls the Agent and its controller. Tool providers are independent actors; the evaluated provider explicitly supplies the capabilities in Table 5, without granting general service control. We consider three observers, distinguished by their rights. A *transit observer* sees connection destinations, directions, ciphertext lengths and times on the links it watches. A *recipient* additionally reads the requests addressed to it, including arguments. An *active auditor* also chooses probes. Identifying a requested action does not establish whether it was authorized; that needs the principal and policy context. Table 2 maps these rights to common deployments and to the existing evidence for each view.

Where cover can terminate. Transport cover ends at an endpoint the defender controls, after the observer and before the application [34]. A relay outside a monitored egress can serve as that endpoint, but the hop from the relay to a third-party service remains. On that hop, the service's own protocol stack still offers two terminators the defender can use without the provider's cooperation. TLS 1.3 lets the sender pad any record and send padding-only records, which the receiver's TLS layer discards [25, Sec. 5.4]; HTTP/2 lets the sender pad DATA and HEADERS frames and send PING frames, which the receiver's HTTP layer consumes [24, Secs. 6.1, 6.7]. These mechanisms can shape the defender's own request bytes and timing. A padding-only record does not request an application response; PING requests a protocol acknowledgement with an eight-octet payload. Neither mechanism by itself specifies an application-shaped response or its service latency.

Cover termination therefore depends on the private signal and the provider interface. Request-side padding leaves application response activity to be addressed separately (Table 3). A cooperating provider can synthesize padded, delayed responses without executing the business operation. Where that interface is unavailable, additional application calls are one way to generate response activity. Our experiment compares calls executed on isolated clones; it is not an impossibility result for all other defenses. A recipient that operates the service sees every exchange it receives, so against it cover must also carry plausible arguments. Reaching the application does not by itself require a business write: a read-only or no-op call elicits a real response. Cover must perform a state-changing operation only if such operations produce responses or latencies that reads cannot imitate. We call any cover that reaches the application *executed cover*; the design space in Section 4 separates how much business work it performs.

Observer in this paper. Our measurements use a transit observer on the final hop to seven tool services, with no defender relay in between, and a privacy target that includes the exchange sequence and response timing. It receives destinations, ciphertext bytes and TLS record completion times. Section 7.1 additionally supplies episode and call windows, an evaluator privilege that Section 7.2 withdraws. The recipient view, where decoy arguments are also visible, is part of the threat model but not of these measurements (Section 8). The adversary may train on traffic generated under the same cover policy; the attack suite therefore includes classifiers and cover-removal models fitted on protected training executions, alongside attacks frozen on unprotected traffic.

4. The Executed-Cover Design Space

Four axes. We classify a cover policy along four axes (Table 4). *Termination* and *state* determine what business work cover performs: a disposable copy, read-only access, an isolated clone or production state. *Feedback visibility* determines whether cover responses enter later model requests and can change the real planner's decisions. The *dispatcher*, the process that decides when cover is sent, determines whether cover inherits the timing of honest calls. Honest calls follow a model decision; a controller that inserts cover without waiting for the model creates a timing signature that the first three axes do not capture. Our live study varies the dispatcher while holding termination (an isolated clone) and feedback (hidden) fixed; an earlier state control varies termination and feedback.

The execution contract. Because executed cover performs work, a policy must satisfy a contract as well as a privacy target. The contract names the objects and transitions cover may touch, the quota, token and latency budgets it may consume, and the feedback it may expose to the real planner. Isolation prevents direct writes to production state but still consumes compute and can change scheduling; a response hidden from the planner does not undo the operation that produced it. A requested goal can complete while another order, a shared quota or a ledger is wrong, so the contract separates three outcomes: requested-goal completion, violations of declared business invariants, and resource use. For a synthetic fixture whose intended transition is fully specified, exact state equality is the invariant; in production, invariants must admit legitimate concurrent updates.

The service contract. Every policy also needs capabilities from the service (Table 5). Our live study gives the defender a *cooperating provider*: services pad response bodies, hold a cover response for a sampled minimum duration, and execute cover on an isolated clone. This matches a provider that offers a privacy mode or a sandbox on the same endpoint, while its network path is observed by others. Such a provider could also discard cover in its application and return padded, delayed dummy responses, which is the proxy-discard row of Table 4; our dispatcher result applies to that design too, because it concerns when cover requests

TABLE 1. EVIDENCE LOCATED IN THE RETAINED RECORDS. COUNTS USE ELIGIBLE SETTINGS; THE SECOND COLUMN ALSO GIVES DISTINCT SOURCES. "NOT LOCATED" LEAVES EVIDENCE UNRESOLVED WITHIN THE INSPECTED SCOPE. THESE COUNTS DESCRIBE AVAILABILITY, NOT STUDY QUALITY OR CHECKLIST VALIDITY.

Question	Eligible settings / sources	Located	Supporting settings and consequence
Protected-condition attack adaptation	4 / 2	2/4	Communication graphs: transformed captures and live OHTTP. Both come from one preprint.
Original-task utility under the same intervention	4 / 2	1/4	WRA visibility comparison. It does not price WRA's executed-decoy condition.
Adaptation and original-task utility in one setting	4 / 2	0/4	Adaptation and utility are located in different records; transport cost alone does not fill the task-outcome field.
Original-task utility for executed cover	1 / 1	0/1	WRA's five research decoys report recovery and extra research; matched original-task utility is not located.

TABLE 2. DEPLOYMENT AND ACCESS MAP. THE CONFIGURATIONS ARE ANALYST-DEFINED; CITED SETTINGS TEST THE LISTED VIEW, NOT EVERY COMPONENT OF THE DEPLOYMENT. A GATEWAY'S PLAINTEXT ACCESS REQUIRES TLS TERMINATION.

Deployment	Observer and permitted view	Existing evidence for that view	Protection consequence
Enterprise egress	A transit monitor sees routed model/tool flows, sizes, times and any exposed destination association.	WRA: supplied research-domain traces; ANT and cloud benchmark: model/combined traffic [2], [9], [4].	Specify which services cross the monitored egress. A union of flows is available only when the monitor can observe them.
Shared gateway or relay	Transit sees multiplexed gateway output. A terminating gateway can additionally read addressed requests.	Live OHTTP: passive-link inference under stated relay trust [3]; PrivacyTrace: semantic recipient/gateway access [35].	Separate relay association protection from content trust. Connection sharing changes the observation unit.
Local model, remote tools	An external observer sees remote tool traffic; the local model call has no corresponding external network leg.	WRA's destination-recovery mechanism is relevant to tool egress [2]; whole-deployment protection is untested in the mapped comparison.	Evaluate tool and downstream calls. Do not import a remote-model attack score into an absent network leg.
Cloud model, remote tools	An authorized vantage may see either or both encrypted legs. A contacted tool sees its own arguments.	ANT/cloud benchmark test model traffic [9], [4]; PrivacyTrace tests recipient-visible representations [35].	Check permitted model and tool views after the intervention; recipients and passive monitors need different protections.

TABLE 3. WHAT EACH TERMINATOR HIDES FROM AN OBSERVER OF THE FINAL HOP TO A THIRD-PARTY SERVICE. PROTOCOL-LEVEL PADDING IS DISCARDED BY THE SERVICE'S TLS OR HTTP/2 LAYER WITHOUT ITS COOPERATION [25], [24], BUT RESPONSES STILL FOLLOW REAL APPLICATION CALLS.

Cover mechanism	Terminated by	Still visible to the observer
Defender tunnel or relay	Defender endpoint	Everything on the relay's upstream hop
TLS record padding, padding-only records	Service TLS layer	Response presence, size and latency of real calls
HTTP/2 frame padding, PING	Service HTTP/2 layer	As above; PING acknowledges an 8-octet payload
Read-only or no-op call	Application	Read-versus-write response differences; quota use
Business operation (sandbox or live)	Application	Only what the attack can still separate

are sent, not what the service does with them. A non-cooperating provider offers none of the service-side capabilities: the observer additionally sees unpadded response sizes and unheld response timing, and cover executes against real state. We therefore read our leakage results as measured under defender-favorable service conditions, and we do not measure the non-cooperating case.

Where executed cover sits among defenses. Name hiding and relays change destination association; padding changes lengths; pacing changes timing [17], [18], [66]. These layers remain useful, and Table 3 shows when transport cover suffices. When response activity is private, cover responses are needed as well as cover request bytes. A cooperating provider can synthesize them at an application terminator; otherwise the service interface determines which calls can produce them (Section 3). Model services add their own signals, such as token streams, cache behavior and token counts [20], [12], [11], [13], [21]; a tool-side cover policy does not shape later model requests. Appendix B summarizes what each defense layer requires the defender to control, the costs to measure together and the exposure that remains.

TABLE 4. COVER TAXONOMY. EACH POLICY IS CLASSIFIED BY WHERE IT TERMINATES AND WHAT STATE IT TOUCHES, WHETHER ITS RESPONSES REACH THE REAL PLANNER, AND WHICH PROCESS DISPATCHES IT. THE LAST COLUMN NAMES THE EVALUATION IN WHICH THE POLICY IS MEASURED; S1 IS THE EARLIER STATE CONTROL AND SECTION 7.1 THE LIVE TIMING STUDY.

Policy	Terminates at / state touched	Feedback to planner	Dispatcher	Measured in
Transport cover	Tunnel endpoint before the service; no business state	No	Sender schedule	STP [34]; WF padding [66], [18]
Proxy discard	Defender proxy computes on a disposable copy	No	Controller	S1
Read cover	Real service; reads only	No	Controller	S1
Isolated execution	Real service on a persistent clone	No / yes	Controller	S1 (hidden and visible); all live cover arms
Live write	Real service on production state	No	Controller	S1 (stress control)
Research decoys	Real third-party Web destinations	Not located	Decoy generator	WRA [2]
Immediate	Isolated clone	No	Controller, directly after an honest response	Section 7.1 (calibration)
History-matched	Isolated clone	No	Controller, training-gap and duration samples; sequential or concurrent	Section 7.1
Planner shadow	Isolated clone; separate planner context	No	Second planner on an independent ticket	Section 7.1

TABLE 5. SERVICE CAPABILITIES USED BY THE LIVE STUDY AND WHO MUST PROVIDE THEM. THE STUDY THEREFORE MODELS A COOPERATING PROVIDER; AGAINST A NON-COOPERATING PROVIDER ONLY CLIENT-SIDE CAPABILITIES REMAIN.

Capability	Used by	Provided by	Non-cooperating provider
Response body padding (16 KiB quanta)	All arms	Service	Unavailable
Server-side hold of a cover response	History arms	Service	Unavailable; client can delay only its next request
Isolated clone for cover execution	All cover arms	Service	Only with a sandbox on the same endpoint
Request padding, dispatch timing	All arms	Client	Available
Same-service serialization	All arms	Client	Available

TABLE 6. JOINT EVALUATION CHECKLIST FOR ONE INTERVENTION AND POPULATION. ATTACH EVIDENCE OR STATE THE MISSING ITEM; GOAL COMPLETION AND WHOLE-STATE CORRECTNESS ARE SEPARATE FIELDS.

☐	Hidden target, population, prior or base rate
☐	Allowed observations and supplied boundaries
☐	Cover termination, privileges and isolation
☐	Planner-visible cover feedback and dispatcher
☐	Declared tolerances: inference, goal, state, tokens, bytes, latency
☐	Adapted attacks and largest tested residual
☐	Original-task attempts, completion and failures
☐	Business invariants and violations
☐	Matched bytes, latency, tokens and quota

5. Evaluation Methodology

5.1. Joint evaluation protocol

Privacy, task success, business state and cost must refer to the same intervention and evaluation population. The protocol has four steps; Table 6 is its reporting form.

1) **Specify observations and execution authority.** Name the hidden target, the visible links, recipient rights and any supplied boundaries. State where cover ends, which state it can access and whether its responses reach the planner.
2) **Specify the contract and tolerances.** Define the requested goal, allowed collateral transitions, and the inference, quota, token and latency tolerances, before measuring.
3) **Measure matched executions and adapted attacks.** Pair baseline and policy on the same workload. Include cover-removal and schedule-aware attacks, report the strongest tested residual inference, and take task, state and cost outcomes from those protected runs.
4) **Assess requirements.** Fix the attack selection and uncertainty procedure before evaluation. Joint acceptance requires simultaneous support for every mandatory endpoint; an unresolved endpoint leaves the policy undecided. Record forbidden executions separately from goal completion and compare observers using the same attack suite. Our worked example is descriptive because full joint uncertainty is not implemented.

TABLE 7. IDEAL CANDIDATE-LABEL REFERENCE $P^*(k)$ FOR k SHADOW PLANNERS AND SIX CLASSES, COMPUTED EXACTLY (EXPERIMENT/SHADOW_FLOOR.PY). IID: SHADOW LABELS DRAWN UNIFORMLY; DISTINCT: DRAWN WITHOUT REPLACEMENT FROM CLASSES OTHER THAN THE REAL ONE. TOKENS ARE NOMINAL PLANNER MULTIPLES. ONLY $k = 1$, IID, WAS RUN.

k	iid	distinct	Tokens×	Live result ($k = 1$)
0	1.000	–	1	
1	.583	.500	2	Max 70 / 64 of 120
2	.491	.333	3	
3	.456	.250	4	
4	.428	.200	5	
Test-split reference: 70.5/120 (21 matching labels)				

5.2. Attacks and statistics

Each condition has its own attack suite, fitted on its training split. Size-only attacks are logistic regression (LR) and a 300-tree random forest (RF) on destinations and ciphertext sizes, a classifier frozen on no-cover traffic, and last- and penultimate-destination rules. Timing attacks add timing features, supervised and label-free gap removal, context-aware removal and joint decoding; Supplement S2.3 defines them. The following conventions apply to every result. *Sel.* is the attack chosen on validation data before test scoring, and primary tests use only these choices. *Max* is the largest test count in a finite suite; it is descriptive and is not an upper bound on inference. *Exploratory* marks an analysis designed after test results were seen. Paired tests are exact McNemar tests with Holm correction within each prespecified family; intervals use 10,000 action-stratified fixture bootstraps and condition on the fitted attacks and designed workload.

5.3. An ideal candidate-label reference

Decoy-query obfuscation hides a query among candidates [67]. For planner cover, each candidate is a complete task. Consider the ideal observation consisting only of the unordered real and k shadow labels, all recovered perfectly. If the real chain is exchangeable with the shadows, its posterior label probability is $n_c/(k+1)$, where n_c counts chains labeled c. The modal-label guess succeeds with probability

$$P^*(k) = \frac{\mathbb{E}[\max_c n_c]}{k+1}.$$

This is optimal accuracy for that ideal label observation. With one shadow and six uniform classes, $P^*(1) = \frac{1}{6} + \frac{5}{6} \cdot \frac{1}{2} = \frac{7}{12}$. Table 7 gives the exact enumeration. Drawing distinct shadows from the other labels yields $1/(k+1)$ while preserving each shadow's uniform marginal.

The six-class prior reference is $1/6$; $P^*(k)$ instead describes ambiguity when all candidate labels are known but roles are hidden. Actual traces may obscure labels or disclose roles, so $P^*(k)$ is neither a universal upper nor lower bound on trace inference. Reused fixtures and cache histories also limit exchangeability.

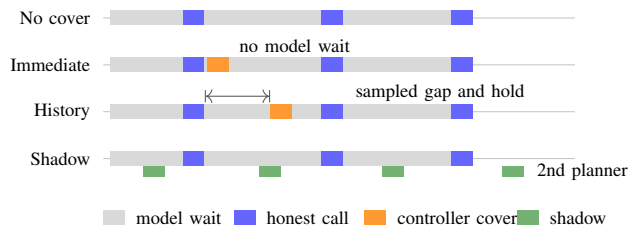


Figure 2. Dispatchers in the live study. Honest calls follow a model wait. Immediate cover follows the previous response without one, which label-free gap removal exploits. History-matched cover samples a training gap and response hold. Shadow calls come from a second planner, so their gaps are model waits too.

6. Experimental Setup

Workload and planners. Each fixture is a synthetic support ticket requesting one of six business actions over a state with three orders, a customer address, a compensation quota, an escalation status and a ledger. LangGraph [56] alternates model decisions with real tool calls to seven MCP Streamable HTTP services over TLS 1.3/HTTP/2, with request and response bodies padded to 4,096 and 16,384 bytes. Two fixed-weight planners, Qwen3:8B and Llama3.1:8B, run at temperature zero with weights pinned by layer hashes. Each planner executes 300 fixtures per condition: 120 training, 60 validation and 120 paired test fixtures, balanced over the six actions. The label is the requested action, and wrong or failed executions remain in every denominator.

Cover policies. All four cover arms execute cover on an isolated clone and hide its responses from the real planner; they implement different call-generation strategies (Figure 2). *Immediate* cover follows an honest response directly. *History* cover samples a training-only honest gap and service duration, waits in real time and holds the response for the sampled minimum; the *sequential* variant blocks the real chain and the *concurrent* variant overlaps real model inference. *Shadow* runs a complete second planner on an independent ticket whose action is drawn uniformly, and a fair draw decides which chain starts first. Calls to the same service serialize. The shadow also changes call count and composition. Controller cover positions are fixed before collection among slots from just after the ticket to just after the reference business action; in 79 of the 120 test fixtures no cover is planned after that action, so unless the planner deviates the requested action remains the final call. All no-cover episodes are collected before the cover arms, whose order rotates across fixtures; prompt caches persist across arms. The five conditions give 3,000 main episodes and 600 shadow chains.

Studies. Table 8 lists the populations. The live timing study and its record-only reanalysis answer the privacy questions. A separate earlier state control on the same workload varies termination and feedback across six policies, including a deliberately harmful live-write arm, to test whether goal completion misses collateral state. All runs use one

TABLE 8. STUDY MAP. EACH POPULATION ANSWERS ONE QUESTION AND KEEPS ITS OWN DENOMINATORS AND STATISTICAL ROLE.

Study	Population; observer	Question; role
Live timing (§7.1)	3,000 episodes + 600 shadow chains; call windows, TLS record times	Does the dispatcher expose cover? Primary families P, R, C
Record-only (§7.2)	Same episodes; no call boundaries	How much survives without supplied boundaries? Retrospective; one primary pair
State control (§7.4; S1)	3,600 episodes; call windows, sizes only	Does goal completion miss collateral state? Primary state family

host over loopback with synthetic tools; Section 8 states what this does not test.

7. Results

7.1. A dispatch shortcut and residual trace context

Immediate cover exposes its short gap. A label-free two-component mixture over training gaps selects the short-gap component for removal, and a classifier trained on no-cover traffic decodes the rest. With supplied call windows it recognizes 112/120 and 108/120 tasks, against selected size-only attacks at 75 and 80 (Table 9). Both gains meet the replication rule after Holm correction ($p = 1.9 \times 10^{-8}$ and 4.3×10^{-6}). Independent TCP reassembly matches all 773,450 TLS record types, lengths and counts, and packet-capture clocks reproduce both scores. The mixture flags 181/182 and 182/182 held-out immediate decoys with no false flags (Supplement S2.7). This directly supports the short-gap mechanism in the measured planner loop.

Sampled waits leave recognizable call context. Sampling training gaps defeats the mixture: recognition falls to 53 and 14 (sequential) and 38 and 66 (concurrent). Context removal or joint decoding recognizes 93/120 and 104/120 sequential tasks against size-only attacks at 76 and 82, and 99/120 and 106/120 concurrent tasks against 66 and 68. Three comparisons survive Holm correction: Llama sequential ($p = .00078$) and both concurrent comparisons ($p < 10^{-4}$); Qwen sequential does not ($p = .0559$).

These attacks use destinations, positions, sizes and timing relations. Direct timing-plus-size classification is weaker: Qwen scores 57 and 47 in the history arms, versus size-only LR's 61 and 52 (Supplement Table S6). Thus these policies leave useful trace context, but the history comparisons do not isolate timestamps. Because most controller schedules leave the requested action last, the last-destination rule alone still recognizes 75, 76 and 66 Qwen tasks in the three controller arms (Supplement Table S6). Whether the adaptive attacks' correct predictions concentrate on fixtures with an uncovered terminal action requires the retained per-fixture predictions and is not reported here. Sampled waits are minimums rather than matched conditional timing laws: Qwen's 40.5 ms mean hold becomes 100.4 ms sequential and 171.2 ms concurrent at the observer (Table 10). Llama's sequential mixture deletes 477/634 honest calls while detecting only 64/182 cover calls, explaining its 14/120 score; joint decoding on the same traces reaches 104/120. A failed removal model therefore leaves a separate residual-recognition question.

7.2. What survives without supplied call windows

We reanalyze all 3,000 episodes from complete encrypted connections, including setup and teardown. The attacker receives relative record times, directions, ciphertext lengths and destinations. Episode association remains supplied, but call endpoints, operation names, roles and HTTP stream assignments are removed. An idle-gap rule proposes events; raw-record classifiers skip this step. A training-only mixture removes short-gap events. The immediate-arm pair is the only primary test, and this suite lacks the stronger context and joint decoders of Section 7.1. Selected removal recognizes 69/120 and 86/120 tasks against baselines of 80 and 90: paired differences -9.2 and -3.3 points (Holm $p = .025$ and $.54$). Thus the prespecified removal pipeline loses its advantage without supplied call windows. Its idle-gap events, however, match true calls with IoU-.5 F1 of 0.000 in eight of ten cells, including both immediate cells (Supplement Table S14), so removal operates on segments that do not correspond to calls. The negative result diagnoses this segmentation, not the protective value of withholding boundaries. A subsequent exploratory eventizer uses the padding quantum and MCP setup sequence. It changes Qwen's removal score from 68 to 91 and Llama's from 90 to 87, with event F1 .295 and .380 at IoU .5. It was designed after the first results and adds no confirmatory superiority claim.

Every arm retains above-chance recognition in the tested suite (Table 11). This supports residual recognition from associated encrypted connections and diagnoses this removal pipeline. Unknown-task discovery and strong context attacks on inferred events are separate questions. Unequal representations and suites prevent a causal comparison of boundary access.

7.3. Recognition and work under planner shadows

The strongest tested call-window attack, a size/destination RF, recognizes 70/120 Qwen and 64/120 Llama shadow tasks; validation-selected attacks score 54 and 64. Record-only maxima are 73 and 69 over 52 candidates, with selected attacks at 69 and 69. These are residuals of the complete shadow policy.

The shadow uses 8.14 and 10.46 total tool calls per fixture against controller cover's 5.6 and 6.8, changes their composition to a complete task, and uses 2.01 and 1.99 times the model tokens. Real-chain time rises from 2.91/3.33 s to 4.80/3.94 s; call-window bytes rise 1.68/1.71 times. This

TABLE 9. JOINT EVALUATION OF THE FIVE LIVE POLICIES; 120 TEST FIXTURES PER ROW, SINGLE HOST OVER LOOPBACK, SYNTHETIC TOOLS. SEL. S/A: VALIDATION-SELECTED SIZE-ONLY / ADAPTIVE ATTACK (PRIMARY). MAX: LARGEST TEST COUNT IN THE FROZEN SUITE (DESCRIPTIVE). GMM: LABEL-FREE REMOVAL. REF.: UNIFORM SIX-CLASS REFERENCE 20/120, OR FOR SHADOW THE IDEAL CANDIDATE-LABEL REFERENCE ON THE REALIZED TEST SPLIT (TABLE 7), NOT A BOUND ON TRACE INFERENCE. BYTES: WHOLE-CONNECTION / CALL-WINDOW FACTORS. TOKENS INCLUDE BOTH PLANNERS. GOAL AND STATE ARE SEPARATE COUNTS FROM THIS COHORT; THE STATE-CONTROL COHORT (SUPPLEMENT S1) HAS ITS OWN COUNTS.

Planner	Policy	Sel. S/A	Max	GMM	Ref.	Bytes×	Tokens×	Goal/State
Qwen3:8B	No cover	117/117	117	–	20	1.00/1.00	1.00	113/114
Qwen3:8B	Immediate	75/112	112	112	20	1.15/1.33	1.00	111/112
Qwen3:8B	History, sequential	76/93	99	53	20	1.15/1.32	1.00	112/113
Qwen3:8B	History, concurrent	66/99	99	38	20	1.15/1.33	1.00	111/112
Qwen3:8B	Planner shadow	54/54	70	34	70.5	1.31/1.68	2.01	112/113
Llama3.1:8B	No cover	110/110	112	–	20	1.00/1.00	1.00	82/70
Llama3.1:8B	Immediate	80/108	108	108	20	1.13/1.27	1.01	83/67
Llama3.1:8B	History, sequential	82/104	104	14	20	1.13/1.26	1.01	84/67
Llama3.1:8B	History, concurrent	68/106	106	66	20	1.13/1.27	1.02	82/67
Llama3.1:8B	Planner shadow	64/64	64	43	70.5	1.35/1.71	1.99	82/67

TABLE 10. ACHIEVED HISTORY-BASED COVER TIMING, ALL SPLITS (455 COVER CALLS PER ROW). MEANS IN SECONDS. AN ANCHOR GAP IS MEASURED FROM THE ACTUAL PRIOR COMPLETED ANCHOR CALL TO COVER RECORD START. SAMPLED DURATIONS ARE MINIMUM SERVER HOLDS, WHILE ACHIEVED DURATIONS INCLUDE RECORD-LEVEL TRANSPORT AND SCHEDULING.

Model	Arm	Drawn gap	Achieved anchor gap	Drawn hold	Achieved duration
Qwen	History seq.	0.5198	0.5660	0.0405	0.1004
Qwen	History concurrent	0.5198	0.5772	0.0405	0.1712
Llama	History seq.	0.5865	0.6203	0.0768	0.1097
Llama	History concurrent	0.5865	0.6204	0.0768	0.1234

TABLE 11. NO SUPPLIED CALL BOUNDARIES; WHOLE-EPISODE ASSOCIATION REMAINS SUPPLIED. B/R: VALIDATION-SELECTED IDLE-GRID BASELINE AND REMOVAL ATTACKS (PRIMARY PAIR: IMMEDIATE). MAX: LARGEST TEST COUNT OVER BOTH SUITES, INCLUDING THE EXPLORATORY PROTOCOL-AWARE EVENTIZER DESIGNED AFTER THE IDLE-GRID RESULTS WERE SEEN (DESCRIPTIVE). COUNTS OUT OF 120; UNIFORM REFERENCE 20.

	Qwen		Llama	
Policy	B/R	Max	B/R	Max
No cover	115/109	117	108/101	112
Immediate	80/69	91	90/86	97
History, sequential	73/74	75	88/56	93
History, concurrent	56/61	62	78/62	88
Planner shadow	69/65	73	69/56	69

quantifies the complete policy's trade-off. Unlike the controller arms, the shadow adds a second business action to every episode, which alone could lower terminal-destination attacks. Isolating online planner generation requires a controller replay with matched calls, composition and terminal coverage.

The ideal observer in Section 5.3 knows both candidate labels but no real-chain identity. With 21 matching pairs among 120 fixtures, its reference is 70.5/120. The measured counts are compatible with that reference, which neither bounds actual trace attacks nor proves role indistinguishability. The chain attack recovers the real chain exactly in only 29 and 32 fixtures. On the 99 different-label cases, aggregate RF counts bound task accuracy only to $[49, 70]/99$ and $[43, 64]/99$; this is not a direct role test. Cache histories and start-order strata motivate that separate analysis (Supplement S2.5–S2.6). The shadow equivalence family compares identical selected attacks and supplies no indistinguishability evidence.

7.4. Business outcomes from the same runs

The live study records goal, state and cost on the same executions as traffic. All cover runs on isolated clones. Goal/state counts stay within three fixtures of no cover (Table 9); honest paths match no cover on 115/120 Qwen and 103/120 Llama shadow fixtures. Llama's baseline goal/state counts, 82/70 of 120, define a limited executor baseline rather than a production reliability target.

A separate state-control cohort tests whether these endpoints distinguish collateral effects. Proxy, read and hidden-isolated cover keep Qwen state correctness at 115–116/120. Deliberate live writes reduce it to 7/120 while 102/120 goals complete, including 95 with collateral errors. Llama has 3/120 intact and 73 of 76 completed tasks with collateral errors. Both integrity reductions survive Holm correction ($p = 3.7 \times 10^{-32}$ and 3.0×10^{-19}; Supplement S1). These constructed cases demonstrate endpoint sensitivity; their separate counts estimate no production damage rate.

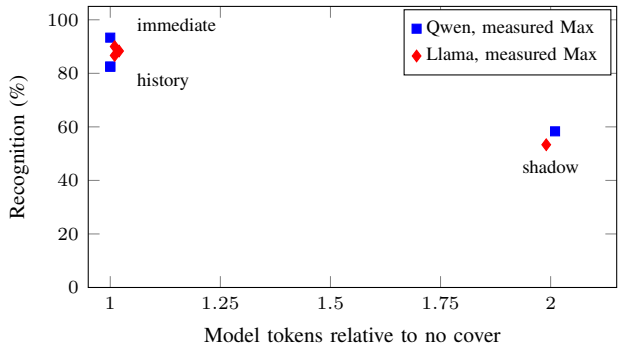


Figure 3. Measured recognition and model work for the four cover policies. Points are strongest tested call-window attacks; overlapping controller points retain their values. Shadow changes call count and composition as well as model work. Ideal candidate-label values are shown separately in Table 7.

7.5. A worked screening example

Table 12 uses illustrative tolerances chosen after measurement. It is a descriptive worksheet, not a validated deployment decision. All call-window controller maxima exceed 60%; shadow's displayed count intervals straddle it. A decision study must fix selection and uncertainty procedures in advance and cover every mandatory outcome, including paired goal/state changes and costs.

The sample has limited resolution near 60%. An attack performing exactly at the iid reference $7/12 \approx .583$ needs about 3,400 fixtures before its 95% Clopper–Pearson upper limit falls below 60% (1,983/3,400 gives .5999); at the distinct-label reference of .5, 60/120 already gives .593. Neither calculation is a powered role-equivalence study. Distinct-label sampling changes the ideal reference and remains unmeasured.

Record-only history results come from a weaker suite and cannot establish protection relative to the call-window suite. Observed mean costs are below the worksheet budgets; complete joint uncertainty is unassessed. Figure 3 shows the measured policy points; Table 7 gives the ideal label references separately.

8. Discussion

What the measurements add. Where an implementer generates a tool call becomes part of its traffic signature. Immediate cover bypasses the model wait and exposes a short-gap feature. Marginal sampled waits remove that shortcut while leaving task context useful to the tested attacks. This directs evaluation toward destinations, ordering, sizes and timing relations together. It also separates a demonstrated timing mechanism from a stronger classifier's mixed-context result.

The shadow policy changes that sequence at a measured cost. Lower recognition and approximately doubled tokens describe a complete policy point; choosing online generation over controller replay requires a matched comparison. Execution authority is a second design choice. A cooperating provider can synthesize a response, use a sandbox, or execute a business operation. Those choices have different state and resource obligations even when they seek similar traces.

Measurements that distinguish the explanations. Timestamp ablations can separate context from timing. Controlled replays can hold calls and composition fixed while changing dispatch; stronger decoders on record-only observations can test reliance on supplied windows. A service-capability comparison can remove response padding and holds while keeping safe synthetic states. A joint model/tool observer can seek whether each tool call follows a model exchange. These are distinct interventions, each requiring its own observation contract.

Scope and uncertainty. This is a single-host study with synthetic tools, two fixed 8B models and cooperating services (Table 5). Same-service calls serialize. Fixtures are paired across conditions and reused across studies; request phrasings recur across splits. Cross-host deployment, unseen templates, unrestricted multiplexing, recipient arguments, non-cooperating services and simultaneous model-link observation are unmeasured; the last two are the common deployment cases. Loopback removes network noise; longer remote inference could strengthen the dispatch contrast. Neither extrapolation is estimated here.

Intervals condition on fitted attacks and reused workloads. Test maxima describe the suite, not a population optimum. The worksheet uses post-measurement thresholds and lacks full joint uncertainty. No-cover episodes were collected first and caches persist across arms, so cover-versus-none comparisons of cost, goal and state include phase effects; Llama's immediate and concurrent real-chain latencies below no cover ($0.88\times$ and $0.84\times$, Table 12) are consistent with such effects and are not read as speedups. Within-arm attack comparisons (families P and R) are unaffected. The separate state stress cohort deliberately permits collateral writes, so it illustrates endpoint sensitivity rather than harm prevalence. Appendix A's four privacy settings come from two sources and motivate the joint record without estimating a field-wide rate.

9. Related Work

Agent side channels. Section 2 and Appendix A cover Agent communication attacks. WRA tests research decoys and destination visibility [2]; the communication-graph study tests transformations and live OHTTP [3]. Serving-side studies examine streams, cache sharing and token counts [20], [12], [11], [13], [21], [14]. Our controlled study follows calls into TLS traces while measuring work under each policy.

Decoy-query obfuscation. TrackMeNot generates periodic queries [63]; GooPIR combines real and decoy candidates [67]. Classifiers and systematic studies show how generated queries remain distinguishable [64], [65], [68]. This generator principle is the starting point here. We examine its interaction with model-decision latency and tool authority

TABLE 12. DESCRIPTIVE SCREENING WORKSHEET. TOLERANCES WERE CHOSEN AFTER MEASUREMENT: RECOGNITION $\leq 60\%$, GOAL/STATE POINT DIFFERENCES AT MOST 3/120 BELOW NO COVER, MODEL TOKENS $\leq 2.5\times$, CALL BYTES $\leq 2\times$ AND MEAN REAL-CHAIN LATENCY $\leq 2\times$. CW MAX IS THE LARGEST TESTED COUNT WITH A NOMINAL 95% CLOPPER–PEARSON COUNT INTERVAL, NOT AN INTERVAL FOR ALL-ATTACKER PERFORMANCE OR A PROSPECTIVELY SELECTED ATTACK. GOAL/STATE AND COSTS ARE DESCRIPTIVE. ABOVE AND UNRESOLVED REFER ONLY TO THE WORKSHEET; NO POLICY HAS A VALIDATED JOINT PASS. RO HISTORY USES A WEAKER SUITE.

Planner	Policy	CW Max [95% CI]	RO Max	$\Delta G/\Delta S$	Tok.$\times$	Bytes$\times$	Lat.$\times$	CW screening	RO screening
Qwen3:8B	Immediate	112 [.87, .97]	91	$-2/-2$	1.00	1.33	1.23	Above: inference	Above: inference
Qwen3:8B	History, sequential	99 [.75, .89]	75	$-1/-1$	1.00	1.32	1.47	Above: inference	Unresolved; suite incomplete
Qwen3:8B	History, concurrent	99 [.75, .89]	62	$-2/-2$	1.00	1.33	1.14	Above: inference	Unresolved; suite incomplete
Qwen3:8B	Planner shadow	70 [.49, .67]	73	$-1/-1$	2.01	1.68	1.65	Unresolved: inference	Unresolved: inference
Llama3.1:8B	Immediate	108 [.83, .95]	97	$+1/-3$	1.01	1.27	0.88	Above: inference	Above: inference
Llama3.1:8B	History, sequential	104 [.79, .92]	93	$+2/-3$	1.01	1.26	1.15	Above: inference	Above: inference
Llama3.1:8B	History, concurrent	106 [.81, .94]	88	$0/-3$	1.02	1.27	0.84	Above: inference	Above: inference
Llama3.1:8B	Planner shadow	64 [.44, .63]	69	$0/-3$	1.99	1.71	1.18	Unresolved; state at point limit	Unresolved: inference

using actual dispatch, two observation representations and matched business/resource outcomes. The candidate-label reference is the deniability calculation for complete task labels.

Traffic defenses and Agent evaluation. WF supplies adapted-attack, observer and overhead requirements [15], [16], [32], [17], [18], [66], [69], [62]; smart-home shaping terminates cover at a tunnel [34]. Agent information-flow controls govern tool authority [57], [58], [59]; SPILLage examines behavioral oversharing [27]. Environment outcomes in τ-bench and AgentDojo [60], [61] motivate our state endpoint. We bind those endpoints to protected traffic, complementing lifecycle and attack-surface surveys [53], [54], [55].

10. Ethics and Availability

Ethics. Inputs are synthetic, captures are restricted to the experiment, and state-stress writes affect isolated fixtures. The hosted pilot preserved in the evidence archive used account-holder authorization; provider research terms remain unverified, and it supports none of the present main results. Traffic inference can enable profiling; our measurements target synthetic tickets. AI tools assisted extraction, implementation and writing; observations and analyses are preserved. Credentials, hosted transcripts and captures, and third-party full texts are excluded from distribution.

Availability. The accompanying review artifact provides manuscript sources, versioned collection and analysis implementations, fixture generation, frozen plans and historical timing pools, aggregate tests and costs, the ideal-reference computation, and the source-located background matrix. Its index distinguishes arithmetic checks from full-data replication. Released aggregates support paired-test arithmetic and reference enumeration. Per-episode records, individual timestamps and predictions, captures and TLS keys remain local. Independent refitting, feature reconstruction and per-fixture role checks require those withheld inputs. No public release of them is promised.

11. Conclusion

Cover generation and execution must be evaluated together. In the measured Agent loop, controller insertion bypasses a model wait and produces a removable short-gap signature when calls are delimited. Sampled waits defeat that shortcut while mixed-context attacks retain task recognition. With inferred events, the prespecified removal pipeline loses its advantage. Complete shadows lower measured recognition at the cost of more calls and model work; the comparison does not isolate online generation. Recording inference, goals, business state and cost on matched executions connects these traffic results to the authority and resources a policy consumes.

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Appendix A.
Background Evidence Scope

The companion matrix contains forty-four settings from fifteen Agent communication studies. Inclusion requires an LLM to select actions with external feedback and a communication-side inference or evaluated protection. Client, model, tool and coordination interfaces are eligible; transit, recipient and active access remain separate. The bounded window covers first public submissions from 1 January 2024 through 9 November 2026. The evidence archive preserves the exact retrieval, screening, source-version and coding record (its background-extraction report, Sections A and S4–S6).

The settings comprise twenty-two transit observations, ten recipient settings, eleven active settings and one inferred-preference setting. Their source-defined scores are background evidence, not a common benchmark. The four query/workflow privacy settings cited in Section 2.4 come from two sources. The source-located matrix and archived coding record permit inspection of that narrow comparison without treating it as a field-wide frequency.

Appendix B.
Defense Layers

The evidence archive (its background-extraction report, Sections B and S7) retains the full layer comparison: naming/association, representation, scheduling, transport cover, execution cover and shared-state isolation. The current study concerns call generation and execution authority; Table 5 states the service capabilities actually used.